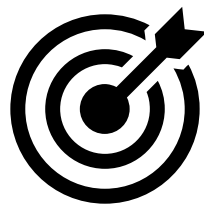


“MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE”



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1. Abstract

The real estate industry generates a significant volume of customer and property transaction data. However, identifying meaningful buyer segments and investment behaviours from such data remains a challenge. This project aims to develop a machine learning-based buyer segmentation framework for real estate market intelligence using customer demographic and transactional information.

The study involves data cleaning, feature engineering, exploratory data analysis, feature encoding, feature scaling, and customer segmentation using the K-Means clustering algorithm. The Elbow Method and Hierarchical Clustering were employed to determine and validate the optimal number of customer segments, while the Silhouette Score was used to evaluate clustering quality. An interactive Tableau dashboard was also developed to visualize buyer demographics, investment behaviour, customer satisfaction, and geographic trends.

The analysis identified three meaningful customer segments with distinct demographic and purchasing characteristics. These insights provide valuable support for customer profiling, targeted marketing, investment planning, and strategic decision-making in the real estate sector.

2. Introduction

The real estate sector is one of the largest contributors to economic development and investment activity worldwide. Real estate organizations collect extensive information regarding customer demographics, purchasing behaviour, financing preferences, and property transactions. Transforming this data into actionable insights has become increasingly important for improving customer targeting, optimizing marketing campaigns, and supporting investment decisions.

Traditional customer analysis often relies on descriptive statistics and manual segmentation, which may fail to uncover hidden behavioural patterns within large datasets. Machine learning techniques, particularly clustering algorithms, enable organizations to identify naturally occurring customer groups based on similarities in demographic, financial, and transactional characteristics.

This project presents a machine learning-based buyer segmentation framework for real estate market intelligence. Python was used for data preprocessing and machine learning, while Tableau was used to create an interactive dashboard for business intelligence. The combination of predictive analytics and data visualization provides meaningful insights that can help real estate organizations understand customer behaviour, improve investment strategies, and enhance customer relationship management.

3. Problem Statement

Real estate companies manage large volumes of customer and transaction data, yet many organizations lack a systematic approach to identifying meaningful customer segments. As a result, marketing campaigns often target broad customer groups rather than specific buyer profiles, reducing campaign effectiveness and limiting business growth. Furthermore, differences in customer demographics, acquisition purpose, geographic location, financing behaviour, and purchasing patterns make manual segmentation difficult. Without data-driven customer profiling, organizations struggle to identify profitable customer groups, recommend appropriate investment opportunities, and improve customer satisfaction. This project addresses these challenges by applying machine learning techniques to segment customers into meaningful groups based on demographic and transactional characteristics. The resulting insights support targeted marketing, customer profiling, and strategic investment planning.

4. Research Objectives

The primary objective of this study is to develop a machine learning-based buyer segmentation model that supports intelligent decision-making in the real estate market.

The specific objectives are:

1. To preprocess and clean the customer and property transaction datasets.
2. To analyse customer demographics and purchasing behaviour through exploratory data analysis.
3. To identify significant patterns in customer acquisition, financing, and investment preferences.
4. To apply K-Means clustering for buyer segmentation.
5. To validate clustering results using the Elbow Method, Hierarchical Clustering, and the Silhouette Score.
6. To interpret customer segments based on demographic and behavioural characteristics.
7. To develop an interactive Tableau dashboard for visualizing customer segments and investment trends.
8. To provide strategic recommendations for improving customer targeting and investment decision-making.

5. Literature Review

Customer segmentation is an important analytical technique used to classify customers into distinct groups based on their demographic characteristics, purchasing behaviour, and financial preferences. Effective segmentation enables organizations to design targeted marketing campaigns, improve customer satisfaction, and allocate business resources more efficiently.

Machine learning has significantly improved customer segmentation by identifying hidden patterns in large datasets without requiring predefined customer categories. Among various clustering algorithms, K-Means is widely adopted because of its computational efficiency and ease of interpretation, while Hierarchical Clustering provides a visual representation of relationships between data points through dendrograms.

In the real estate industry, buyer profiling helps organizations understand differences in investment behaviour, financing preferences, geographic distribution, and purchasing intentions. Traditional segmentation methods often rely on simple demographic analysis, whereas machine learning techniques reveal complex behavioural relationships that support more informed business decisions.

Although previous studies have demonstrated the effectiveness of clustering techniques for customer segmentation, many have focused on retail, banking, or e-commerce datasets. Comparatively fewer studies integrate customer demographics, financing behaviour, geographic information, and property transaction details into a single buyer segmentation framework for real estate market intelligence. This project addresses that gap by combining exploratory data analysis, machine

learning, and interactive business intelligence dashboards to provide actionable insights for customer profiling and investment planning.

6. Dataset Description

The dataset used in this project consists of customer demographic information and real estate transaction records. Customer details and property information were merged into a unified dataset to facilitate comprehensive analysis and buyer segmentation.

The final dataset contained 7,305 customer transaction records and included demographic, financial, behavioural, and property-related variables.

Feature	Description
client_id	Unique customer identifier
client_type	Unique customer identifier
Gender	Country of residence
Country	Country of residence
Region	Geographic region
date_of_birth	Customer date of birth
acquisition_purpose	

	Home purchase or Investment
loan_applied	Loan application status
referral_channel	Customer acquisition source
satisfaction_score	Customer satisfaction rating
listing_id	Property identifier
unit_category	Property category
floor_area_sqft	Property size
sale_price	Property transaction value
listing_status	Property status

Table: Dataset Features

The dataset was used for exploratory data analysis, customer profiling, machine learning-based segmentation, and dashboard development.

7. Research Methodology

The research methodology followed a structured data science workflow comprising data preprocessing, exploratory analysis, machine learning, and business intelligence visualization.

1. Data Collection and Integration

Customer and property datasets were imported into Python and merged using a common customer reference field to create a consolidated analytical dataset.

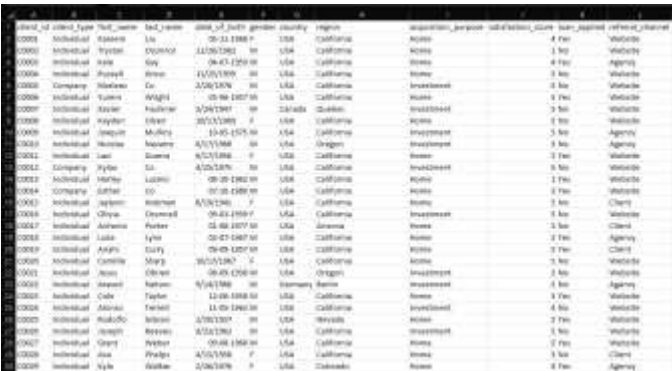


Figure 1.1: Customer Demographic Dataset (clients.xlsx)



Figure 2: Cleaned and Merged Real Estate Dataset

```
merged = pd.merge(
    clients,
    properties,
    left_on="client_id",
    right_on="client_ref",
    how="inner"
)
```

Figure 1.3: Integration of Customer and Property Datasets Using Python

2. Data Cleaning

Data cleaning included:

- Handling missing values.
- Removing duplicate records.
- Converting data types.
- Cleaning numerical variables.
- Creating derived features.

A new Age variable was calculated from the customer date of birth to improve the quality of the machine learning model.

Figure 2: Cleaned and Merged Real Estate Dataset

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted using Python and Tableau to understand customer behaviour and property transaction patterns.

The analysis included:

- **Buyer Type Distribution.**
- **Gender Analysis.**
- **Country Analysis.**
- **Region Analysis.**
- **Acquisition Purpose.**
- **Loan Application Behaviour.**
- **Referral Channel Analysis.**
- **Unit Category Analysis.**
- **Sale Price Distribution.**
- **Customer Satisfaction Analysis.**

Interactive dashboards were created to facilitate business interpretation.

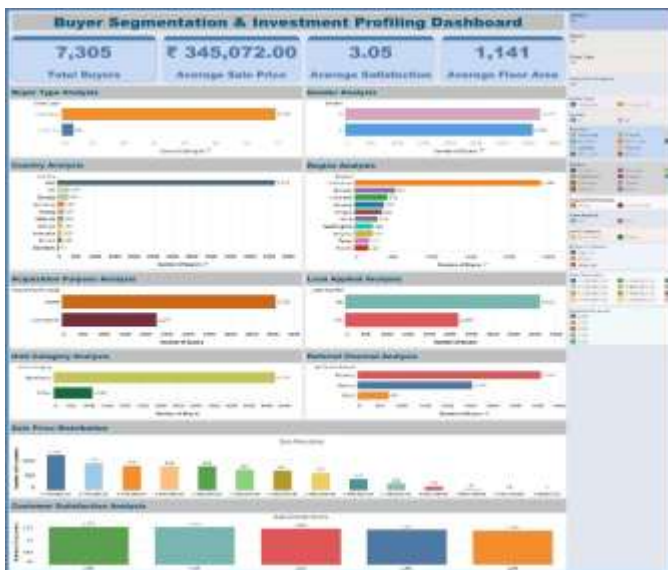


Figure 3: Tableau Dashboard

Relevant features were selected for clustering after excluding identifiers and non-informative attributes.

Feature engineering included:

Age calculation

Feature selection

One-Hot Encoding

StandardScaler normalization

These preprocessing steps ensured that all variables contributed appropriately to the clustering algorithm.

5. Machine Learning

Customer segmentation was performed using the K-Means clustering algorithm.

The following evaluation techniques were applied:

Elbow Method

Silhouette Score

Hierarchical Clustering

The Elbow Method suggested an optimal cluster count of three, while the Silhouette Score (0.302) indicated acceptable cluster quality. Hierarchical Clustering further validated the overall segmentation structure.

4. Feature Engineering

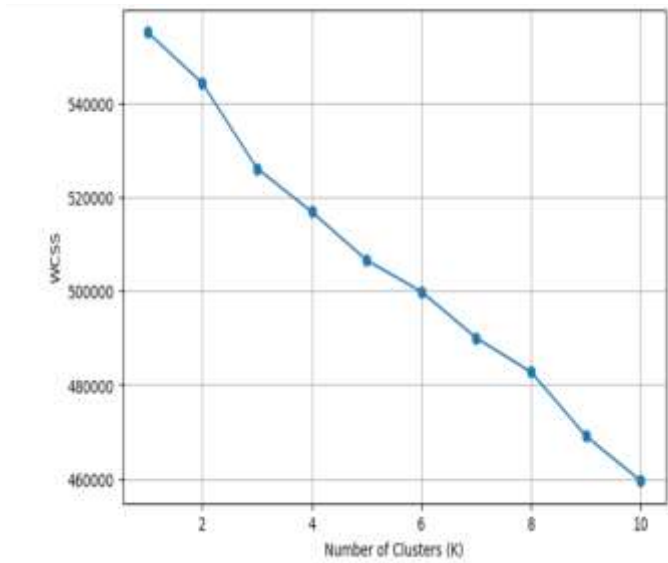


Figure 4: Elbow Method graph and Dendrogram

8. Exploratory Data Analysis (EDA)

The purpose of exploratory data analysis was to identify trends, customer characteristics, and purchasing behaviour before applying machine learning techniques.

8.1 Buyer Type Analysis

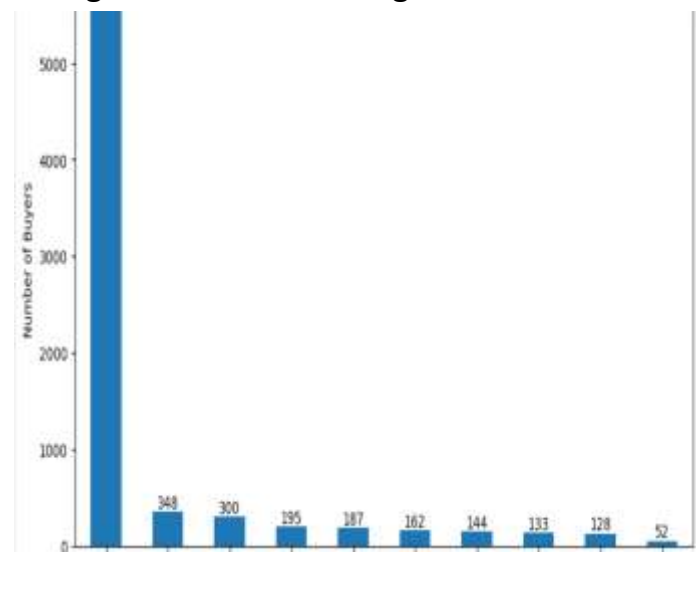
The analysis revealed that individual buyers constitute the majority of customers, whereas company buyers represent only a small proportion of the dataset. This indicates that the real estate market is primarily driven by retail

customers.



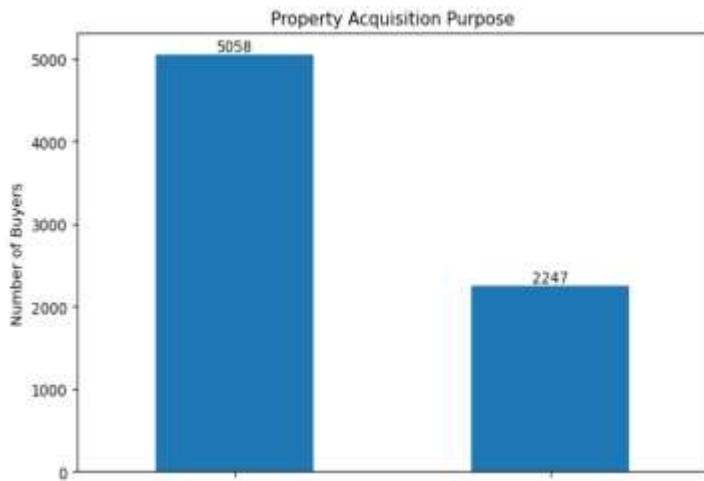
8.2 Country Analysis

The analysis showed that the United States contributes the largest customer base, followed by several international markets. This highlights the importance of regional market strategies.



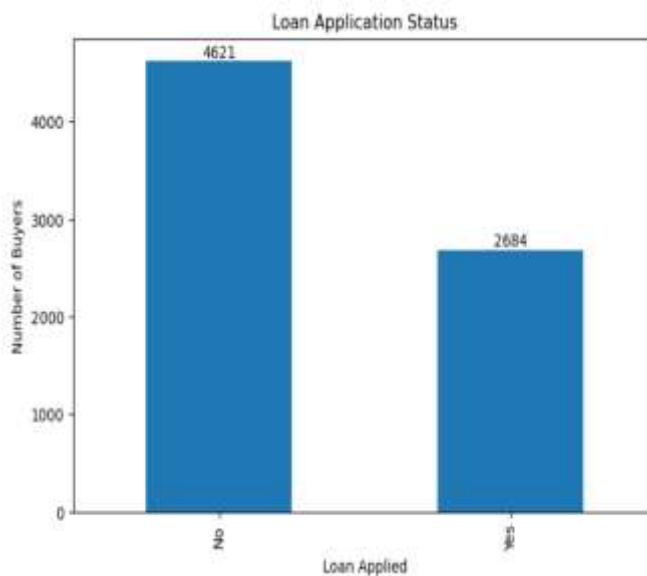
8.3 Acquisition Purpose Analysis

Most customers purchase properties for home ownership, while a smaller segment purchases properties for investment Purposes.



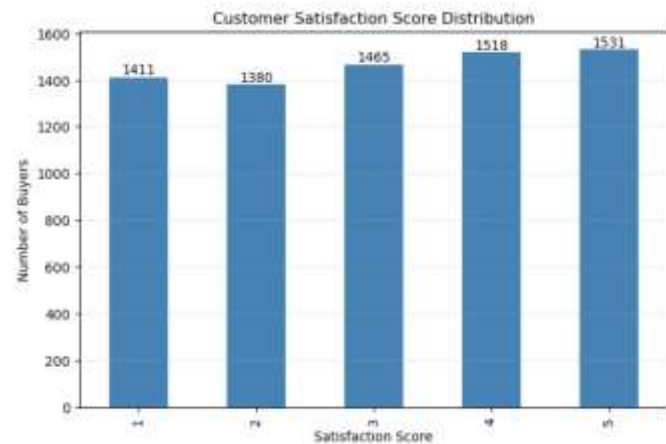
8.4 Loan Behaviour Analysis

The majority of customers purchased properties without applying for loans, indicating relatively strong purchasing capacity among buyers.



8.5 Customer Satisfaction

Customer satisfaction scores were generally moderate to high, indicating positive customer experiences.



8.6 Sale Price Distribution

Most transactions were concentrated within the medium price range, suggesting stronger demand for moderately priced properties than for luxury properties.



9. Machine Learning Results

The K-Means clustering algorithm was applied to identify groups of buyers with similar demographic and purchasing characteristics.

The Elbow Method identified **three clusters** as the optimal segmentation solution. The resulting model achieved a **Silhouette Score of 0.302**, indicating acceptable separation between customer groups.

Hierarchical Clustering was subsequently performed using Ward linkage, and the resulting dendrogram supported the existence of three primary customer groups.

Cluster Summary:

Cluster	Business Name	Characteristics
Cluster 0	International Individual Buyers	Buyers from multiple countries, mostly individual

Cluster	Business Name	Characteristics
		customers, mixed residential and investment purchases
Cluster 1	Canadian Home Buyers	Small cluster, all Canadian, predominantly home buyers with slightly higher satisfaction
Cluster 2	Domestic Residential Buyers	Largest cluster, predominantly USA buyers, mainly residential purchases

These customer segments provide actionable insights for designing targeted marketing campaigns and personalized investment strategies.

11. Results and Discussion

The objective of this study was to segment real estate buyers based on demographic characteristics, purchasing behaviour, and property transaction details using machine learning techniques. The analysis integrated exploratory data analysis, K-Means clustering, Hierarchical Clustering, and interactive Tableau visualizations to generate actionable business insights.

The Elbow Method identified **three clusters** as the optimal number of buyer segments. This was further validated using Hierarchical Clustering, whose dendrogram also indicated three major groups. The clustering model achieved a **Silhouette Score of 0.302**, indicating an acceptable level of cluster separation for a dataset containing multiple categorical and numerical variables.

The cluster profiling revealed distinct customer characteristics:

- **Cluster 0 – International Individual Buyers:** This segment mainly consists of individual buyers from multiple countries with a mix of residential and investment objectives. Their average property purchase value is moderate, making them suitable for personalized investment offers and international marketing campaigns.
- **Cluster 1 – Canadian Home Buyers:** This is the smallest customer segment

and is composed entirely of buyers from Canada. Most customers in this cluster purchase properties for residential purposes and exhibit slightly higher satisfaction levels. Their consistent buying behaviour makes them a valuable niche market for premium residential offerings.

- **Cluster 2 – Domestic Residential Buyers:** This is the largest segment and is predominantly made up of buyers from the United States. Most purchases are for home ownership, and the segment contributes the highest transaction volume. Since this cluster represents the core customer base, organizations should prioritize customer retention, loyalty programs, and personalized communication strategies.

The Tableau dashboard complemented the machine learning analysis by providing interactive visualizations of buyer demographics, geographic distribution, acquisition purpose, loan behaviour, and customer satisfaction. Together, these findings enable stakeholders to better understand customer preferences and support data-driven decision-making.

12. Key Findings

Based on the analysis, the following key findings were observed:

1. The customer base is dominated by **individual buyers**, while company buyers represent only a small proportion of the dataset.
2. The **United States** accounts for the largest share of customers, indicating its importance as the primary target market.
3. Most customers purchase properties for **home ownership**, while investment purchases form a smaller but important segment.
4. A significant proportion of buyer's complete transactions without applying for loans, suggesting relatively strong purchasing capacity.
5. Customer satisfaction scores remain generally positive across all segments, indicating a satisfactory customer experience.
6. K-Means clustering successfully identified **three meaningful customer groups** with distinct demographic and behavioural characteristics.
7. The clustering model achieved a **Silhouette Score of 0.302**, indicating acceptable clustering quality for the available dataset.
8. The Tableau dashboard provides an effective business intelligence tool for monitoring customer trends, segmentation, and investment behaviour.

13. Strategic Business Recommendations

Based on the findings from exploratory data analysis, customer segmentation, and machine learning, the following recommendations are proposed to enhance decision-making, customer engagement, and business performance in the real estate sector.

1. Implement Cluster-Based Marketing Campaigns

The three identified customer segments exhibit distinct demographic and purchasing behaviours. Rather than using a uniform marketing strategy, organizations should design customized campaigns for each segment.

- **International Individual Buyers:** Promote investment-oriented properties, rental income opportunities, and premium developments through digital marketing and international real estate platforms.
- **Canadian Home Buyers:** Focus on residential projects, family-friendly communities, and long-term ownership benefits.
- **Domestic Residential Buyers:** Strengthen customer loyalty through referral programs, exclusive offers, and after-sales engagement.

2. Enhance Customer Relationship Management (CRM)

The segmented customer database can be integrated into CRM systems to deliver personalized communication, recommend relevant properties, and improve customer retention through targeted engagement.

3. Strengthen Geographic Expansion Strategies

Since the majority of buyers are concentrated in the United States, organizations should continue strengthening this market while exploring growth opportunities in underrepresented regions through region-specific marketing initiatives.

4. Improve Investment Advisory Services

Customers purchasing properties for investment require detailed information regarding return on investment, rental income potential, capital appreciation, and market trends. Providing personalized investment insights can improve customer confidence and increase investment-related sales.

5. Optimize Financing Solutions

Loan application analysis indicates varying financing preferences among buyers. Financial institutions and developers can collaborate to offer flexible loan packages, competitive interest rates, and simplified approval processes for eligible customers.

6. Utilize Predictive Analytics

The current clustering model provides descriptive insights. Organizations can extend this work by implementing predictive machine learning models to forecast future buyer behaviour, investment preferences, and customer lifetime value.

7. Develop Real-Time Business Intelligence Dashboards

The Tableau dashboard developed in this study should be regularly updated with new customer and transaction data, enabling management to monitor market trends, customer behaviour, and sales performance in real time.

8. Continuous Model Improvement

As customer behaviour evolves, organizations should periodically retrain clustering models using updated datasets to ensure that customer segments remain accurate and relevant.

14. Conclusion

This study successfully developed a machine learning-based buyer segmentation framework for real estate market intelligence by integrating customer demographic information with property transaction data. The project followed a systematic analytical process,

including data collection, data cleaning, exploratory data analysis, feature engineering, feature encoding, feature scaling, customer segmentation using K-Means clustering, cluster validation through the Elbow Method and Hierarchical Clustering, and business intelligence visualization using Tableau.

The clustering model identified **three distinct buyer segments**, each demonstrating unique demographic and purchasing characteristics. The obtained **Silhouette Score of 0.302** indicates an acceptable clustering structure for the given dataset. The cluster profiles provide valuable insights into customer behaviour, acquisition purpose, geographic distribution, and purchasing patterns.

The Tableau dashboard further enhanced the interpretability of the findings by enabling interactive exploration of customer demographics, property transactions, and investment trends. These visualizations support data-driven decision-making and facilitate effective communication of analytical results.

Overall, the project demonstrates how machine learning and business intelligence techniques can assist real estate organizations in customer profiling, targeted marketing, investment planning, and strategic business development. The developed framework provides a scalable foundation for future analytical applications within the real estate industry.

